INCH-POUND

MIL-T-82786(OS)
5 February 1990
SUPERSEDING
WS 20629
25 October 1983

MILITARY SPECIFICATION

TAPE, CARBON CLOTH, PHENOLIC-PREIMPREGNATED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification defines one type of carbon cloth tape cut from phenolic-preimpregnated carbon cloth broadgoods, referred to herein as the tape.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-B-82792	Broadgoods, Carbon Cloth, Phenolic Pre-impregnated
MIL-T-82846	Thread, Splicing, Nylon

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFRs)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Material.** The tape rolls shall be cut from phenolic-preimpregnated carbon cloth broadgood rolls meeting the requirements of MIL-B-82792 with a bias angle of 45 ± 5 degrees and butt spliced by sewing with nylon thread conforming to MIL-T-82846.

3.3 **Tape widths.** Tape widths shall be in one of four sizes, each ± 1/8-inch, as follows: 2, 2 1/2, 3, or 4 inches.

3.3.1 **Inside and outside diameters.** The tape shall be wound on cores with an inside diameter of 3.0 inches. The outside diameter of the tape rolls shall not exceed 20 inches.

3.4 **Shelf life.** The tape specified herein shall have a shelf life of 9 months after date of manufacture when stored at 50 degrees Fahrenheit (°F) maximum in sealed watertight plastic bags. The cumulative time out of storage of the tape shall not exceed 30 days.

3.4.1 **Shelf life extension.** The shelf life extension of all tape rolls within a single storage container may be extended for one additional 6-month period provided the average value of three specimens from the beginning of one tape roll with the longest out of storage time from the storage container conforms to the following when tested as specified in MIL-B-82792:

	Minimum	Maximum
Resin flow, percent by weight	5.0	25.0
Volatile content, percent by weight	2.5	6.5

3.5 **Workmanship.** The workmanship in stitching shall be such that continuity is ensured throughout the tape roll. The tape shall be free from holes, tears, and other weave anomalies, be uniform in appearance, and free from dirt, grease, oil, and all other foreign materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the quality conformance inspection specified in 4.4. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the inspections and tests specified in table I.

TABLE I. Quality conformance inspection.

Tests and Inspections	Requirement	Test Method
Tape material, width and workmanship	3.2, 3.3, and 3.5	4.5
Packaging	Section 5	4.5

4.5 Visual examination. Each tape roll shall be visually examined for conformance to sections 3 and 5.

4.6 Inspection of packaging. The tape containers and container markings shall be visually examined for compliance with section 5.

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5. PACKAGING

5.1 Packaging. Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 Level C. The tape shall be packaged in watertight plastic bags to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.4. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 Marking.

5.2.1 Standard marking. In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 Special marking. In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Maximum storage temperature (50°F, see 3.4).

d. Acceptance stock number, lot number, and number of rolls.

e. Shelf life and inspection/test date for extension. In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR ONE ADDITIONAL 6 MONTH PERIOD PROVIDED THE REQUIREMENTS LISTED IN 3.4.1 of MIL-T-82786 ARE MET."

5.2.3 Inside roll marking. The inside of each tape roll spool shall be marked with the following, using stamp, stencil, or adhesive label:

a. Manufacturer's name, lot number, material name, and this specification number and revision letter.

b. Stock, lot, and roll number.

c. Date of manufacture.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This carbon cloth tape is intended for use in the manufacture of Standard Missile Mk 104 Dual Thrust Rocket Motor nozzles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Whether first article inspection is required (see 3.1 and 6.3).

d. Width of tape required (see 3.3).

e. Special marking, if other than as specified in see 5.2.2.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors concerning the size of the first article sample and from where it should be taken. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 1340-N793)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-82786	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
TAPE, CARBON CLOTH, PHENOLIC-PREIMPREGNATED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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